



Whitepaper V1

Actionable DeFi Intelligence Layer

0) The Problem

SIGNAL is a decentralized protocol that transforms trading forecasts (“signals”) into **verifiable, monetizable and executable** on-chain strategies.

It introduces:



BlindSignals™

cryptographically time-locked forecasts to prevent copycats and frontrunning.



AlphaScore

a multi-domain reputation system that measures prediction skill across assets and horizons.



SignalVaults

non-custodial smart-contract vaults that route capital to a creator's signals with automated execution and settlement.



MetaVaults

composable portfolios that aggregate multiple creators and strategies under risk constraints.

The protocol targets **crypto, macro, RWAs, commodities and FX** from day one, serving creators (forecasters/strategists), allocators (retail, DAOs, funds), and integrators (protocols, brokers, APIs).

The \$SIGNAL token secures incentives (staking/slashing), grants premium access, pays fees, and governs parameters.

1) Design Goals & Principles



Full transparency

Every forecast, stake, result, and payout is recorded on-chain or backed by verifiable proof.

01



Protection against frontrunning

Encrypted commits and delayed reveals keep forecasts safe until it's time to act.

02



Fast monetization for creators

Earn immediately from unlock fees, subscriptions, and performance fees from vault profits.

03



Built for integration

Open SDKs and APIs let other platforms plug into SIGNAL's execution engine and data feeds.

04



Market versatility

Works with different prediction types: yes/no, price targets, ranges, or relative performance.

05



Safety first

Audited code, kill switches, multiple data sources, and strict risk limits.

06

2) Actors



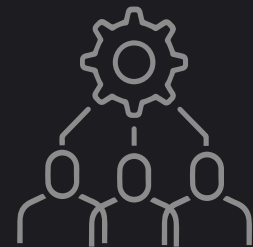
Creators

Publish forecasts, stake their own funds for credibility, earn fees, and can run SignalVaults.



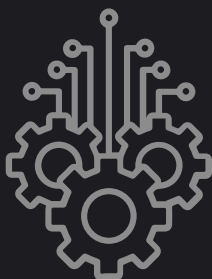
Subscribers

Pay to unlock forecasts, or invest directly via vaults.



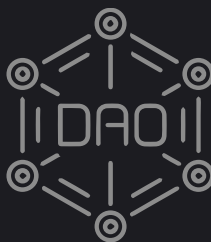
Allocators (DAOs/ Funds)

Deploy larger amounts of capital into vaults or MetaVault portfolios.



Integrators

Use SIGNAL's data feeds or trading engine in their own products.



Governance (DAO)

Decides rules, fees, and treasury spending.



Oracles/Attestors

Provide verified results for predictions and market events.

3) Economic Primitives

3.1 Signal Types

Binary

Yes/No prediction — e.g., price goes up or down, event happens or doesn't.

Target

Predicting that the price will hit a specific level within a set number of days.

Range

Predicting the price will end up between two set levels at a certain time.

Relative

Predicting one asset will perform better than another by a certain percentage in a given time.

Event-driven:

Predicting one asset will perform better than another by a certain percentage in a given time..

3.2 Fees & Payouts



Unlock Fee

When someone wants to see a signal, they pay a fee — this can be a fixed price or adjust based on demand.



SIGNAL

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Performance Fee (Vaults)

Vault managers earn a percentage of profits (e.g., 20%), but only on new gains above the highest value the vault has ever reached (the "high-water mark").



Protocol Share

SIGNAL takes a cut from unlock fees and performance fees. This goes to the treasury, and part of it is used to buy back and burn tokens or reward users.

4) Protocol Architecture

4.1 On-Chain Contracts (EVM, L2 first)

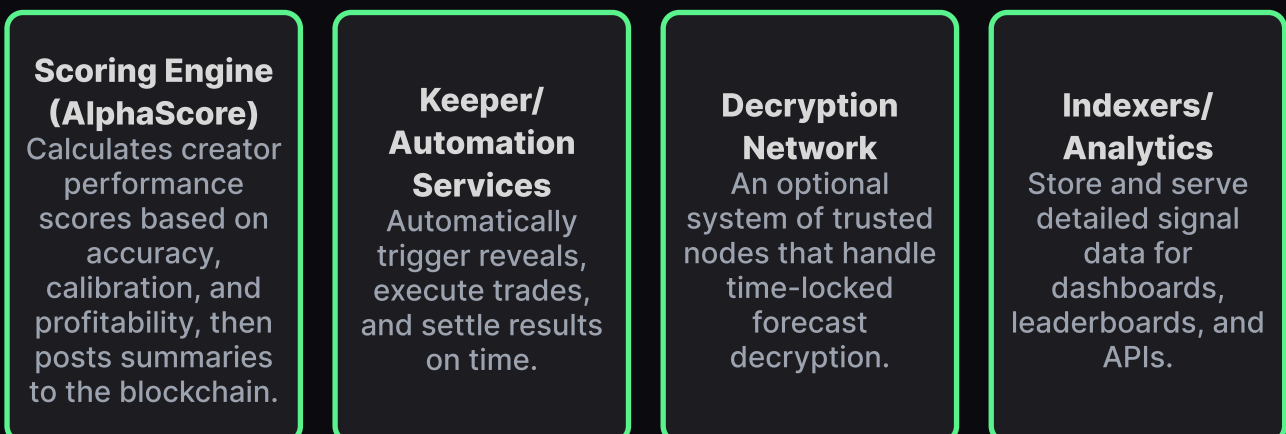
SIGNAL's core logic runs on smart contracts, starting on a low-cost, high-liquidity Layer 2 blockchain like Base or Arbitrum, so transactions are fast and cheap. The key modules are:



Cross-chain execution is possible via messaging systems like Chainlink CCIP or LayerZero. Oracles such as Chainlink and custom attestors provide secure price and event data.

4.2 Off-Chain Services

Some parts of SIGNAL work best off-chain, supported by automation and data infrastructure:



5) BlindSignals™ (anti-frontrun)

5.1 Commit–Reveal (baseline)

- 1 Creator Prepares Prediction**
They decide the details — direction (up/down), target price, time frame, confidence level — and add a secret code (salt) to hide it.
- 2 Commit**
They lock in the hidden version of the prediction on-chain with a set reveal deadline and put up a stake as a deposit.
- 3 Sell/Unlock**
Users can pay to get a sealed, partial description of the prediction without giving away the full details too early.
- 4 Reveal**
Before the deadline, the creator publishes the full prediction and the secret code. The system checks it matches the original hidden version.
- 5 If They Don't Reveal**
Part of their stake is taken away, and users who paid get refunds.

5.2 Extra protection for High-Value Signals

- The prediction is locked using a one-time encryption key. Pieces of this key are given to a trusted group (like Chainlink or Threshold Network). They only release their pieces after a set time or when a specific condition is met.
- This way, not even the creator can reveal the prediction early. It's an optional feature for especially valuable or sensitive signals.

5.3 Private Unlock Receipts

- Unlock buyers receive a **receipt NFT** granting decrypt rights after reveal; preserves privacy for purchasers.

6) AlphaScore (reputation)

Goal: A single, comparable measure of forecasting skill across domains & horizons, robust to gaming.

6.1 Metrics (per signal)



Binary Predictions (Yes/No)

We check how close the predicted probability was to the actual outcome. If you said “80% chance” and it happened, that’s good; if it didn’t, that’s bad. We use math formulas (Brier score or Log loss) to measure this accuracy.



Target or Range Predictions

We measure how close your prediction was to the real final price, using standard “distance from the truth” metrics.



Trade Returns

If your signal runs as a trade, we look at profit adjusted for risk — using Sharpe or Sortino ratios so high, steady returns score better than risky, volatile ones.



Calibration

We check if your confidence levels match reality. For example, if you say “70% chance” often, those things should happen about 70% of the time.

6.2 Rolling Score (per creator, per domain, per horizon)

Each creator gets a score that updates over time based on how good their signals are. Recent signals, bigger stakes, and signals that move more money count more. Older signals slowly lose weight so the score reflects current skill. We turn the raw score into a 0–100 scale using a smoothing formula, and also adjust it like a chess rating (Elo) by comparing creators’ performance against each other.

6.3 Anti-Gaming

We actively protect the system from manipulation:

- Check that forecasts are realistic in size and length.
- Penalize overconfident predictions that turn out to be wrong.
- Use stake-weighted scoring so creators with more at risk have more impact.
- Detect and penalize duplicates or copied forecasts using fuzzy matching.

7) Execution Layer

7.1 Signal → Trade Mapping

Each type of forecast maps to a trade plan



Binary up/down

Could be executed as an options spread or a spot position with a stop-loss.



Target/Range

Placed as ladder limit orders, possibly via batch auctions.



Relative performance

Open a long position on one asset and short on another using perps or spot baskets.

7.2 ExecutionRouter

- The ExecutionRouter is the part of SIGNAL that actually carries out trades. It can connect to different trading platforms like Uniswap, 0x, 1inch, CowSwap, GMX, or perpetual DEXs.
- It follows safety rules — such as limits on price slippage, acceptable price ranges, trade size caps, and gas usage.
- To avoid getting front-run by bots (MEV), it can use “intent settlement,” where trades are bundled and executed in a safer way.
- For big trades, it supports breaking them into smaller pieces over time (TWAP — Time-Weighted Average Price).
- Each vault also has built-in risk limits, like maximum leverage, approved assets, and how big any single position can be

7.3 SignalVault

- A SignalVault is like a smart investment pool. People can put money in or take money out, with a short waiting period for withdrawals.
- It only follows live, revealed signals from the vault’s creator — no hidden or expired calls.
- Profits are shared using a fair “high-water mark” system, so fees are only charged on new gains. These fees are automatically split between the creator and the SIGNAL protocol.
- Built-in safety switches (“guardian circuit breakers”) can pause trading if markets swing too wildly or if there’s a problem with price data.

7.4 MetaVault

We combine several SignalVaults into one portfolio, giving each a certain percentage of the total money. Every so often, we rebalance — adjusting those percentages — while accounting for trading costs.

An optimizer decides the mix that should give the best returns for the level of risk we're willing to take, while making sure the total always adds up to 100% and stays within set limits. We can also use advanced techniques like "risk parity" (spreading risk evenly) or "Black-Litterman" (blending our own views with market data) to fine-tune the portfolio.

7.5 SignalVault vs MetaVault

7.5.1 SignalVault – One Creator, One Strategy

Think of a SignalVault as a dedicated "trading bot" smart contract for a single signal creator.

- People can deposit capital into it.
- It **only follows that creator's active and revealed signals**.
- Trades are executed automatically according to the creator's forecasts.
- Performance fees are shared between the creator and the protocol.
- It's like investing directly into one person's "hedge fund" strategy.

7.5.2 MetaVault – A Portfolio of Multiple SignalVaults

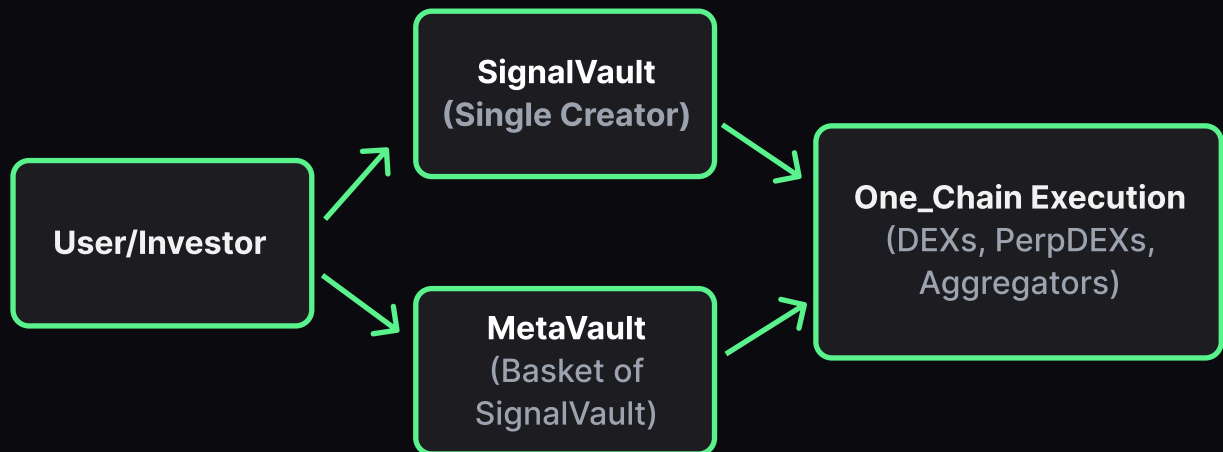
A MetaVault is a portfolio that combines several SignalVaults into one investment.

- Instead of betting on a single creator, it spreads capital across multiple creators/strategies.
- It rebalances holdings periodically based on performance, risk, or market conditions.
- Risk rules (like max exposure to one strategy or asset) are applied at the portfolio level.
- It's like buying an **ETF of trading strategies** instead of picking just one.

Key Difference:

SignalVault = single strategy (tied to one creator)

MetaVault = basket of strategies (multiple creators, risk-managed)



8) Oracles & Outcome Resolution

Price Oracles

We use trusted data sources like Chainlink or DataStream to get asset prices, with safeguards like TWAP (average pricing over time) and requiring agreement from more than one source.

Event Oracles

For things like “company defaulted” or “Fed held rates,” a group of trusted signers confirm the outcome. There’s a short window where anyone can challenge their decision, and these signers have money locked up as a guarantee of honesty.

RWA Oracles

Special feeds provide data on real-world assets, like interest rates, loan defaults, or the value of tokenized funds.

Finalization

Once the dispute window closes, the system locks in the official result, scores the prediction, and sends payouts.

9) Security Model

How We Keep SIGNAL Safe

- 1 Creators Lying or Not Revealing**
If a creator cheats or fails to reveal a signal, part of their stake is taken away and they face penalties.
- 2 Copycats or Front-Running**
We hide signal details until the right time using BlindSignals, delayed reveals, and private unlock receipts.
- 3 Oracle Manipulation**
We use multiple data sources, price limits, and emergency stop mechanisms to prevent fake or tampered prices.
- MEV Attacks**
We protect trades from being front-run by using intent-based settlement (like CoW), breaking big trades into smaller ones (TWAP), setting maximum slippage, and using private transaction relays.
- 5 Vault Blow-Ups**
We limit leverage, only allow approved assets, cap position sizes, and can pause trading in emergencies.
- 6 Fake Accounts (Sybil Attacks)**
Posting is gated by a stake, with limits per ID and checks for linked accounts.
- 7 Spam**
Creators pay a small posting fee that's refunded if their signal is revealed properly.
- 8 Smart Contract Bugs**
All core contracts are audited, the most important math is formally verified, and we run bug bounty programs.

10) Token (\$SIGNAL)

10.1 Utility

Creators stake it to post forecasts (skin in the game).

Users can pay fees in it, and some protocol fees are settled in \$SIGNAL.

Token holders govern parameters, listings, and treasury usage.

Grants access to premium creators and early vault allocations.

Powers AlphaScore data licensing.



10.2 Supply & Allocations (fixed 1,000,000,000)

Token Allocation Distribution

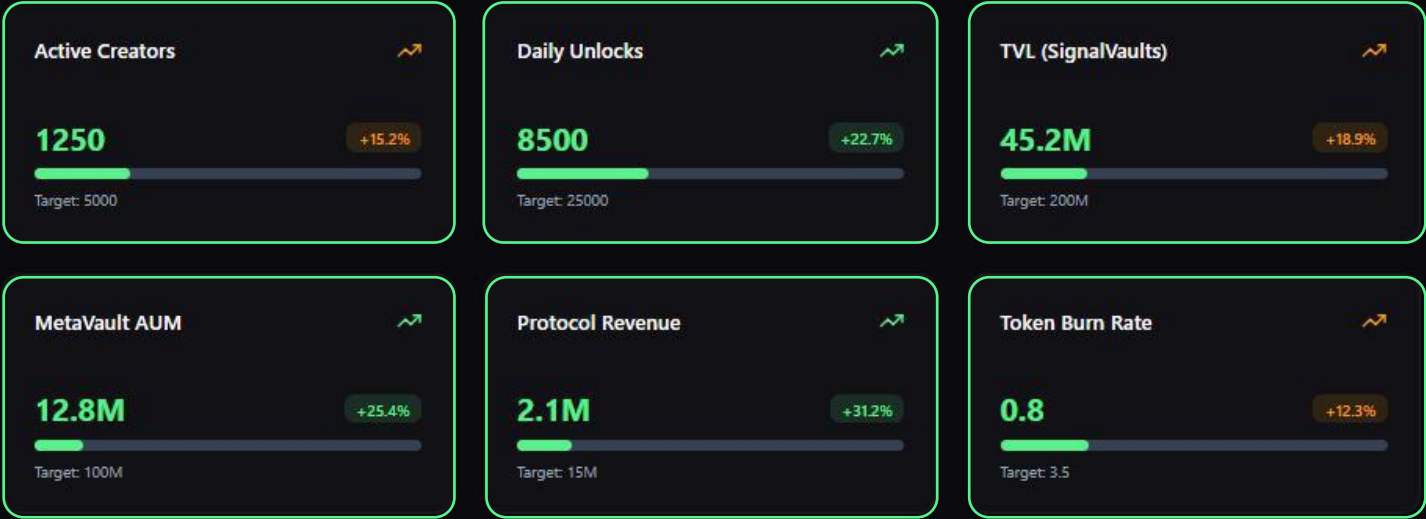


Token Unlock Schedule

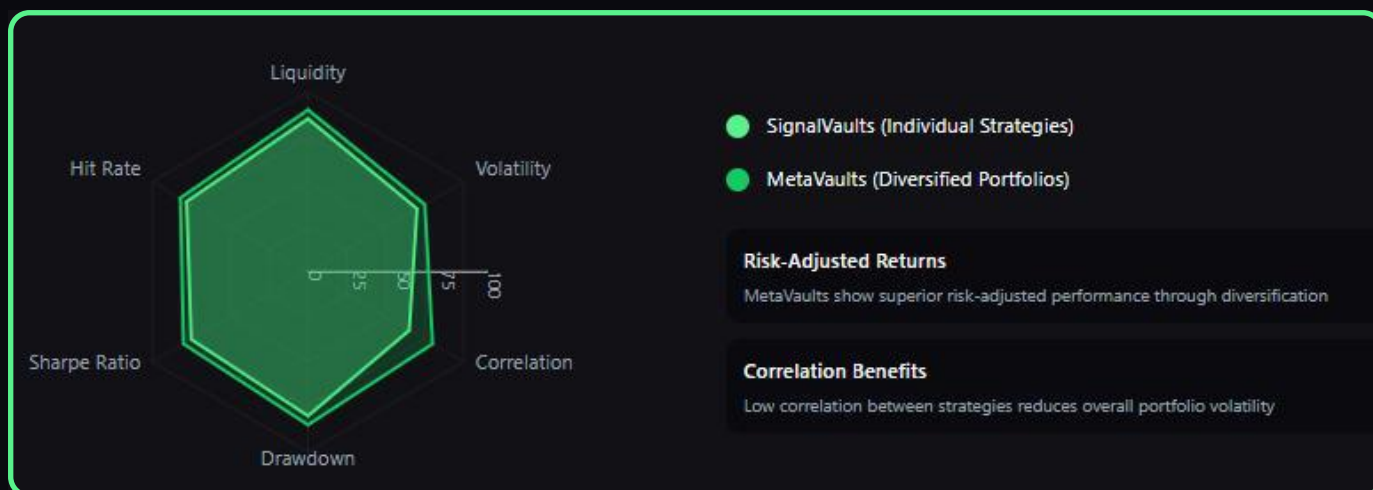


Cumulative token unlock schedule showing gradual release across all categories over 4 years

Performance Analytics (Example)



Risk Profile Analysis (Example)



Max Drawdown

12.5%

VaR (95%)

8.2%

Volatility

24.7%

Beta vs Market

0.68

10.3 Emissions & Sinks

Tokens are emitted based on real, verified performance.

Tokens are “burned” or removed from supply via fee payments, premium access costs, and buyback programs.

Long-term stakers get multipliers on rewards.

11) Governance

We start with a multisig for safety, then transition to a DAO where token holders vote.

Over time, governance becomes modular and fully on-chain, with parameter limits enforced automatically and emergency veto powers for major risks.

12) Compliance & Risk Notice (non-legal)

- All signals are research only — not financial advice.
- Vaults are automated and carry market risk.
- RWA modules may require KYC/whitelisting.
- Geofencing where needed to comply with regulations.
- Clear disclosures on risks and fees.

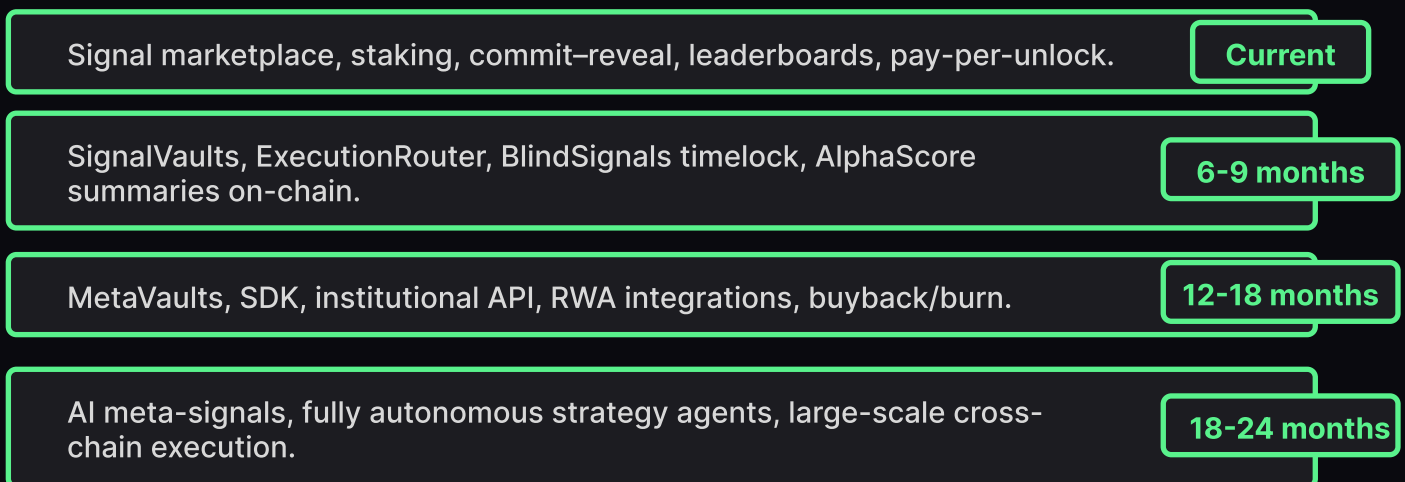
SIGNAL offers both on-chain events and off-chain APIs for developers to integrate signals, vault data, and AlphaScores into other platforms.

13) Data, APIs & SDK

On-Chain Events (selected)

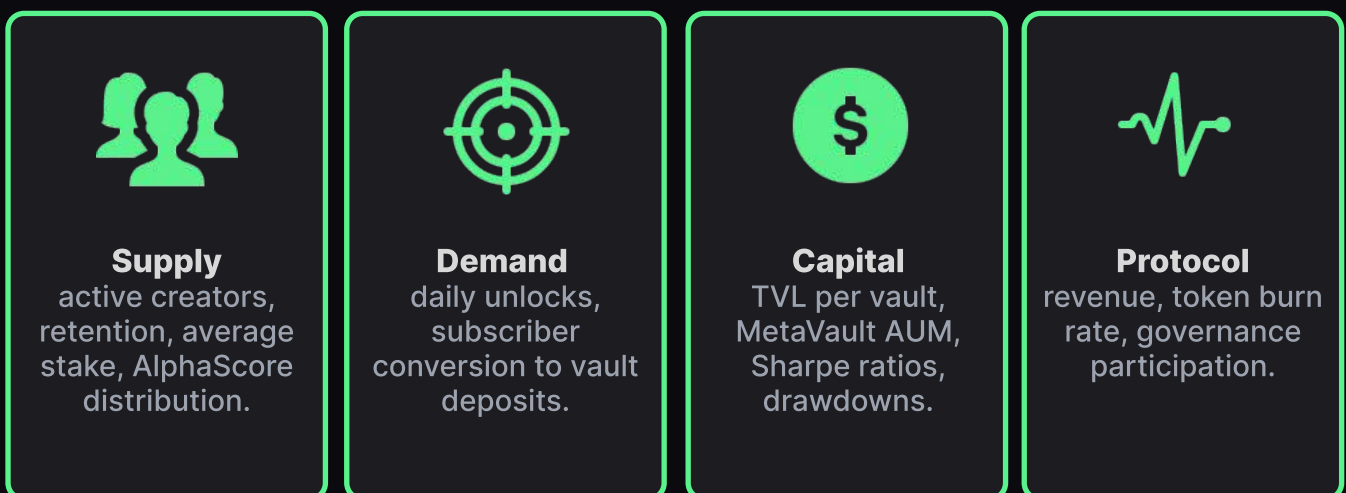
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14) Roadmap



15) KPIs

We measure



16) Appendices

A) Example Signal Payload (hashed)

This is what a signal looks like **before it's revealed** — it's stored in an encrypted form so no one can see the details too early.

B) Scoring Weights (default)

The AlphaScore system uses a weighted formula to judge creators fairly

- 1 Recency Decay**
Newer signals count more than older ones, with influence halving every **90 days**.
- 2 Stake Weight**
Creators who risk more capital get more weight in scoring. The impact grows with stake size but not linearly (diminishing returns).
- 3 Liquidity Weight**
Signals on higher-volume assets get slightly more weight.
- 4 Domain Split**
Final score is a mix of performance in different markets:
 - **40% Crypto**
 - **30% Macro**
 - **30% RWA / Credit**

C) Vault Risk Defaults

SignalVaults have built-in safety settings to protect investor capital

**Max Slippage
per Trade**

0.3%

(prevents bad execution during volatile moves)

Max Leverage

2×

(perpetuals disabled for conservative vaults by default)

**Per-Asset
Exposure Cap**
No more than

25%

of the vault in a single asset

Stop-Loss Rule

If price moves more than 2 standard deviations against the position's recent volatility, the position is closed to limit losses.

17) Funding Plan (Lean)

\$0.5M pre-seed

(SAFE)

complete V2 build (Vaults, Router, BlindSignals)

Milestone follow-on \$3–5M

post-traction to scale
MetaVaults, RWA
integrations, and audits.

TGE after V2 traction

circulating 10–12%; private
allocation to strategic
partners with 6m cliff.

18) Conclusion

SIGNAL makes alpha **verifiable**, **protects it** at the moment of highest value with BlindSignals, and **routes capital** trustlessly through SignalVaults and MetaVaults.

The combination of **reputation**, **execution**, and **composability** positions SIGNAL to become the default on-chain layer where forecasts turn into deployable strategies for retail, DAOs, and institutions.